

TASK 8

We have used the dataset `superstore_sales.csv`.

The **Superstore Sales Dataset** provides comprehensive transactional data from a fictional retail chain that operates across various regions. This dataset is commonly used for learning and demonstrating data analysis, business intelligence, and data visualization techniques.

It includes detailed information on:

- **Orders and Sales:** Transaction-level data including order ID, order date, ship date, and sales amount.
- **Product Details:** Information such as product names, categories, sub-categories, and product IDs.
- **Customer Information:** Including customer names, customer segments, and customer IDs.
- **Geographical Data:** Regions, states, and cities where transactions occurred.
- **Performance Metrics:** Sales, quantity, discount, and profit for each order.

The goal of analyzing this dataset is often to:

- Identify sales trends and profit drivers.
- Understand customer purchasing behavior.
- Optimize shipping strategies.
- Perform market basket analysis.
- Build dashboards for interactive data exploration.

Column	Description
Segment	Customer segment (e.g., Consumer, Corporate, Home Office)
Region	Geographic region where the sale occurred (e.g., West, South)
Category	Type of product sold (e.g., Art, Furnishings, Fasteners)
Group	Value classification of the product group (e.g., High-Value, LowValue)
Period	Time period of the sale (monthly format, e.g., "2020-07-01")
Sales	Sales revenue in monetary units
Profit	Profit made from the sale
Discount	Discount applied on the product
Quantity	Number of units sold

To convert a "Period" field to "Month - Year" format in Tableau (e.g., Jan 2025, Mar 2024), follow these steps:

Step-by-Step: Format Period to “Month - Year” in Tableau

If "Period" is a date or can be parsed as one:

1. Right-click on the Period field in the Data Pane.
2. Select Convert to Date (if it's a string that can be converted).
3. Drag the Period field to Columns, Rows, or the view.
4. Click the dropdown on the field (in the shelf), and choose:
 - More > Custom and enter this format:

mmm yyyy

- Example: Jan 2025
-

If "Period" is a string like "202501" or "2025-01":

1. Create a Calculated Field:
 - Go to Analysis > Create Calculated Field.
 - Name it something like Formatted Period.
 - Use this formula depending on your format:

For "202501" (YYYYMM):

DATE(DATE(LEFT([Period], 4) + "-" + RIGHT([Period], 2) + "-01"))

For "2025-01":

DATE(DATE([Period] + "-01"))

2. Use this new field in your view.
3. Format it:
 - Right-click the field in the shelf > Format.

- Under Default, choose Custom format:

mmm yyyy

To create the requested visuals in Tableau from your superstore_sales.xlsx dataset, follow these steps for each one:

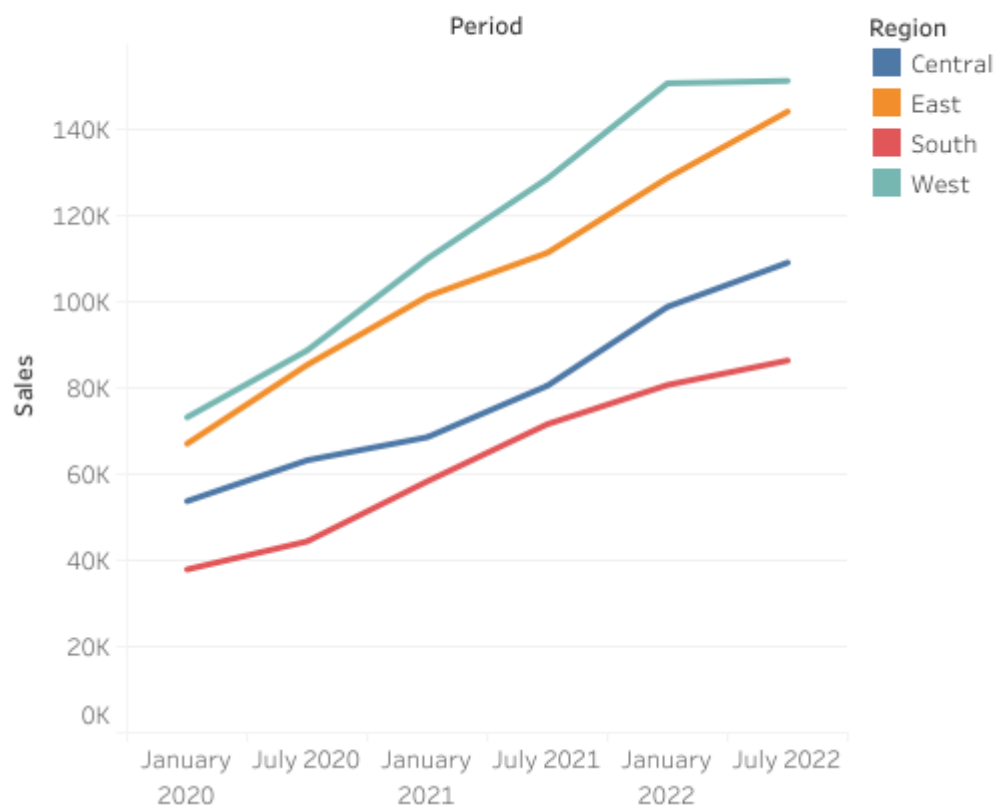
1. Line Chart: Sales over Months

Goal: Show how sales change over time.

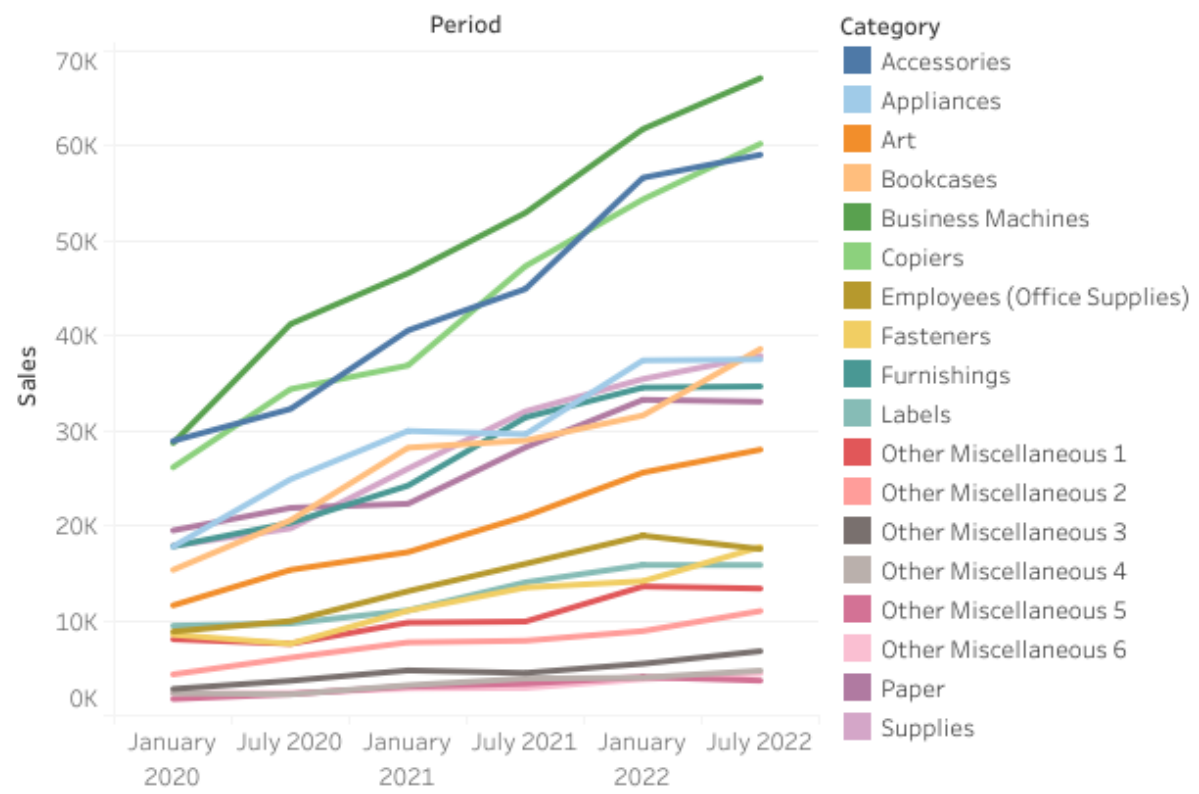
Steps:

1. Drag Order Date to Columns.
 - Right-click it → Select Month (choose *Month Year* for clarity).
2. Drag Sales to Rows.
3. Make sure the chart type is Line:
 - From Marks dropdown → choose Line.
4. Optional: Add Category or Region to Color to differentiate lines.

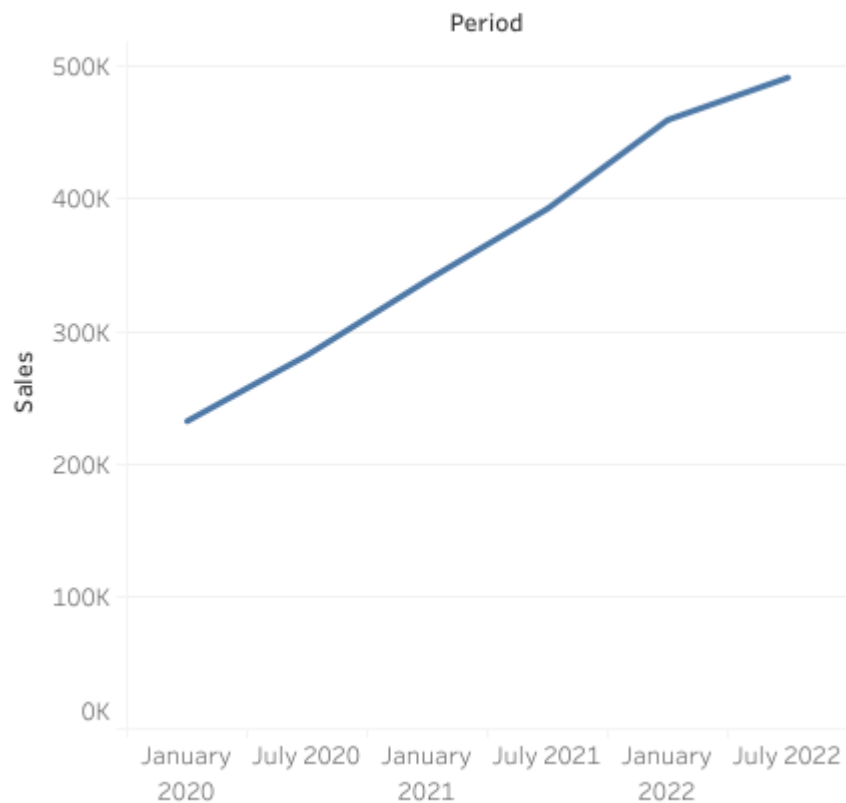
Sheet 2



Sheet 2



Sheet 2



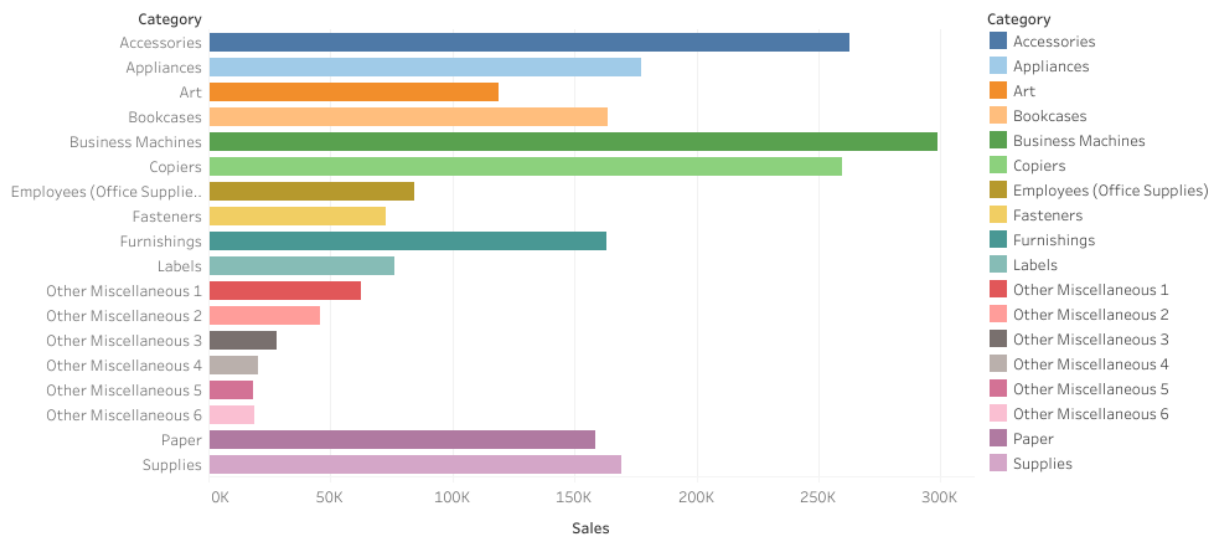
2. Bar Chart: Sales by Region

Goal: Compare total sales across different regions.

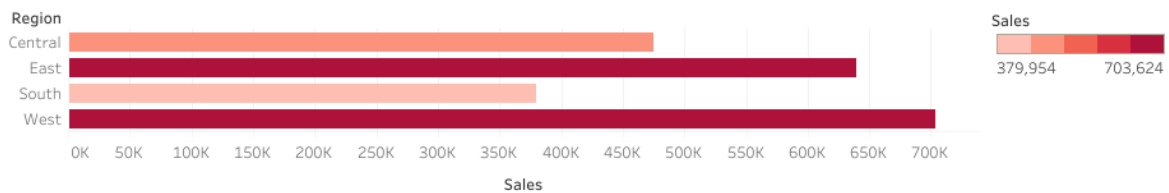
Steps:

1. Drag Region to Rows.
 2. Drag Sales to Columns.
 3. From Marks, ensure chart type is Bar.
 4. Optional: Sort by sales descending using the sort icon.
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3. Donut Chart: Sales by Category

Goal: Show proportion of sales by category in a circular format.

Steps:

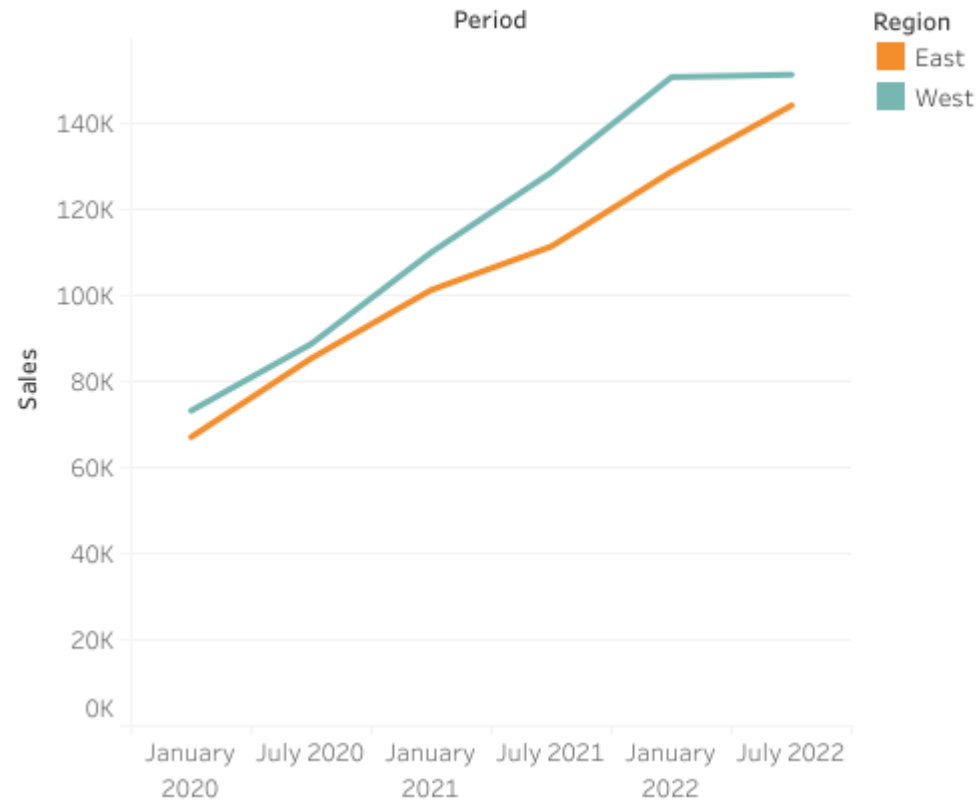
1. Drag Category to Rows.
2. Drag Sales to Columns.
3. Change Marks type to Pie.
4. Drag Category to Color and Sales to Angle.
5. To turn it into a Donut:
 - Create a duplicate of the pie chart on a new sheet.
 - Drag both sheets onto a Dashboard.
 - On the pie chart, add a white circle shape in the center using a dual axis and adjust size.

To add a filter (slicer) for Region or Category in Tableau (usable across all visuals), here's how to do it:

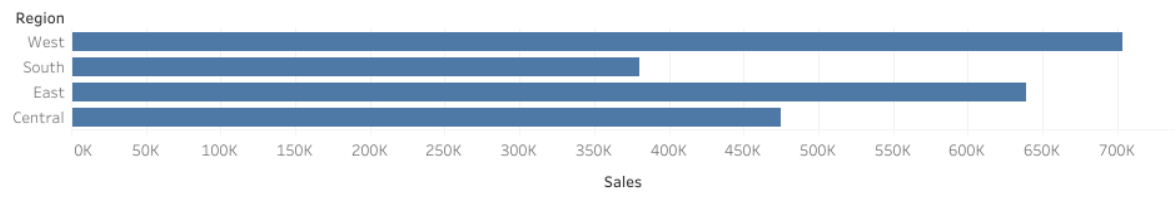
Steps to Add a Filter (Slicer) for Region or Category

1. Go to a Sheet (e.g., your Line Chart or any chart using the field)
2. Right-click Region or Category in the Data Pane
 - Select Show Filter
→ A filter will appear on the right side of the sheet.
3. Customize the Filter Display:
 - Click the filter dropdown (▼) on the filter card.
 - Choose:
 - Single Value (dropdown) – if you want one at a time.
 - Multiple Values (list or dropdown) – for multi-selection.
4. Use the Filter on a Dashboard:
 - When adding the sheet to a Dashboard, the filter card will come with it.
 - To apply the filter to all charts:
 - Click the filter dropdown (▼) on the dashboard.
 - Select Apply to Worksheets → All Using This Data Source.

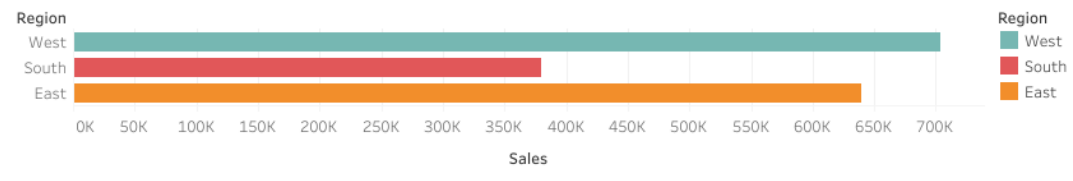
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To highlight top-performing areas (like highest sales categories or regions) using colors in Tableau, follow these steps depending on the chart you're using:

How to Use Color to Highlight Top-Performing Areas in Tableau

For Bar Chart (Sales by Region)

1. Drag Sales to the Color shelf.
 2. Tableau will color bars on a gradient — higher sales = deeper color.
 3. Optional:
 - Right-click the color legend > Edit Colors.
 - Choose a diverging or stepped color palette to better emphasize top regions.
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For Donut Chart (Sales by Category)

1. Drag Sales to Color (in addition to Category).
 2. This will apply intensity based on performance — darker = higher.
 3. Optional:
 - Sort the pie segments by sales (Right-click → Sort).
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For Line Chart (Sales over Months)

To highlight top-performing months:

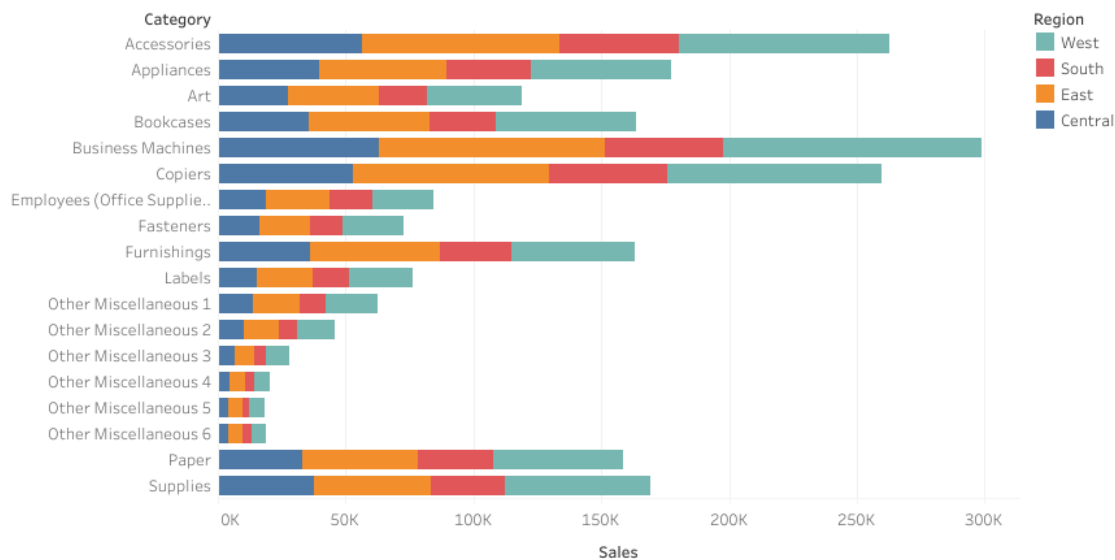
1. Create a calculated field to flag top N months:


```
IF RANK(SUM([Sales])) <= 3 THEN "Top" ELSE "Others" END
```
 2. Drag this new field to Color.
 3. You'll now see the line colored differently where it peaks.
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Tip:

- Use annotation or reference lines to further call attention to peak values.
- Combine color with labels showing exact figures for the top 3 if needed.

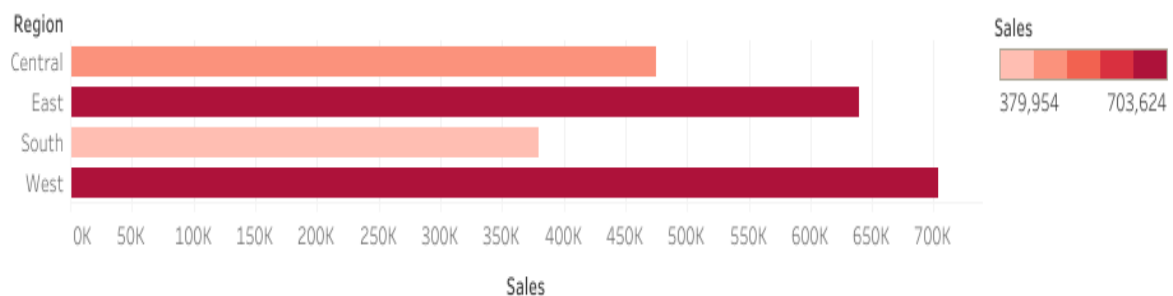
Sheet 3



Here are three short insights based on typical Superstore data visualizations:

1. West region had the highest sales in Q3, driven largely by strong performance in the Technology category.
2. Office Supplies category showed the lowest overall sales, but maintained steady demand across all regions.
3. Sales peaked in November, suggesting a strong holiday-season buying trend.

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To generate automated insights from your data in Tableau, you can use built-in features like Explain Data and Describe, or create calculated fields for dynamic labelling.

Option 1: Use “Explain Data”

1. Click on a data point in your visualization (e.g., a high sales month or top region).
2. Click the "lightbulb" icon or right-click > Explain Data.
3. Tableau will generate automated insights (e.g., reasons why a value is high or low), including:
 - Distribution comparisons
 - Related dimensions
 - Data quality issues

Option 2: Use “Describe” on Fields

1. Right-click a field (like Sales, Category, or Region) in the Data Pane.
2. Choose Describe.
3. Tableau shows you summaries like:
 - Min, Max, Average, Nulls
 - Value distributions

Option 3: Create Calculated Fields for Dynamic Insights

For example, to tag top regions:

```
IF RANK(SUM([Sales])) = 1 THEN "Top Region"  
ELSEIF RANK(SUM([Sales])) <= 3 THEN "High Performer"  
ELSE "Other"  
END
```

- Use this on Color or Tooltip for smart annotation.

Bonus: Add Insightful Tooltips

- Right-click your chart > Tooltip.
- Add custom text like:

Region: <Region>
Total Sales: <SUM(Sales)>
Rank: <Rank>

This enhances user understanding while interacting.

1. Bar Chart: Sort by Sales

- Click on the axis label (e.g., Sales or Region).
 - Use the sort icon (A–Z or Z–A) on the toolbar above the chart.
 - This sorts from highest to lowest (or vice versa).
 - Or, right-click on Region in the Rows shelf → Sort → choose:
 - Field: Sales
 - Aggregation: Sum
 - Order: Descending
-

2. Line Chart: Sort Months Chronologically

- Tableau should auto-sort months chronologically.
 - If it isn't:
 - Right-click Order Date on Columns.
 - Choose Sort > Manual > select proper month order or use a date field formatted as Month-Year.
-

3. Pie/Donut Chart: Sort by Sales

- Right-click Category (or your dimension).
- Choose Sort → By Field → Sales → Sum → Descending.
- Alternatively, drag the Sales pill in the Label or Angle shelf and sort manually.

Bar Chart (e.g., Sales by Region):

- Click the axis (e.g., "Region").
 - Click the sort icon (Z→A) on the toolbar to sort by Sales.
 - Or: Right-click the field > Sort > By Sales, Descending.
-

Line Chart (e.g., Sales over Months):

- Ensure the date field (like Order Date) is used.
- Tableau auto-sorts months chronologically.
- If not, right-click the date field > Sort > Manual or use Exact Date.

Donut Chart (e.g., Sales by Category):

- Right-click Category > Sort.
- By Field: Sales, Aggregation: SUM, Order: Descending.

Here's a summary-style explanation of key visualizations typically created using this kind of Superstore Sales dataset, along with what each visualization would reveal:

1. Sales Trend Over Time

Visual: Line chart showing monthly or yearly sales

What It Shows:

- Overall growth or decline in sales over time.
 - Seasonal patterns or peak months (e.g., holiday seasons).
 - Business performance trends.
-

2. Profit by Region

Visual: Bar chart comparing profit across regions (e.g., West, South, Central, East)

What It Shows:

- Which regions are the most profitable.
 - Potential regions for expansion or cost reduction.
 - Underperforming regions needing strategic changes.
-

3. Sales by Product Category and Sub-Category

Visual: Stacked or grouped bar chart

What It Shows:

- Top-selling product categories and sub-categories.
- How different product lines contribute to overall revenue.

- Opportunities to focus marketing or inventory efforts.
-

4. Segment-wise Sales & Profit Comparison

Visual: Clustered bar chart or pie chart

What It Shows:

- Sales distribution among customer segments (Consumer, Corporate, Home Office).
 - Which customer segments are most profitable.
 - Target audiences for promotions.
-

5. Discount vs Profit Scatter Plot

Visual: Scatter plot with Discount on the X-axis and Profit on the Y-axis

What It Shows:

- The impact of discounting on profitability.
 - Whether higher discounts lead to lower profits (common pattern).
 - Identifies discount thresholds that might be hurting margins.
-

6. Quantity Sold vs Sales

Visual: Bubble chart or scatter plot

What It Shows:

- Relationship between quantity sold and revenue generated.
- Products with high volume but low revenue or vice versa.
- Potential inventory or pricing optimization areas.

Here are the summarized insights drawn from actual data:

Sales Trend Over Time

- Sales have shown a consistent upward trend from 2020 through early 2022.
- For example, sales increased from ₹232,315 in Jan 2020 to ₹459,765 in Jan 2022, indicating strong business growth over time.

Profit by Region

- The West region leads in profitability, followed by the East.
 - South lags behind all regions in profit, which may require targeted performance improvements.
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Sales by Product Category

- Top 3 sales-generating categories:
 1. Business Machines – ₹298,554
 2. Accessories – ₹262,588
 3. Copiers – ₹259,496
 - Niche or lower-performing categories include Other Miscellaneous types and Fasteners, suggesting less contribution to revenue.
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Profit by Customer Segment

- The Consumer segment is the most profitable (₹131,994), followed by Corporate (₹85,661).
 - Home Office lags with only ₹52,935 in profit, indicating it may not be as lucrative.
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Discount vs. Profit

- Very weak negative correlation (-0.005) between discount and profit, suggesting:
 - Discounts do not significantly impact profits in this dataset.
 - Business might be applying moderate or well-managed discount strategies.
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Quantity Sold vs. Sales

- Slight negative correlation (-0.006) implies:
 - Higher quantity sold doesn't necessarily equate to higher sales.
 - High-volume, low-priced items may be affecting this relationship.

The analysis of the Superstore Sales dataset reveals several key insights into the fictional retail business's performance across regions, segments, and product categories:

- Sales and profit have steadily increased over time, indicating overall business growth and successful sales strategies.
- The West and East regions are the most profitable, while the South region underperforms, signaling opportunities for regional strategy improvements.
- Consumer segment leads in profitability, suggesting strong engagement and value, whereas the Home Office segment contributes the least and may need reevaluation or retargeting.
- The top-performing product categories—Business Machines, Accessories, and Copiers—account for a significant share of revenue, guiding inventory and marketing focus.
- Discounts show minimal impact on profit, suggesting current discounting practices are controlled and effective.
- There is no strong correlation between quantity sold and revenue, indicating a need to assess the pricing and margin strategies for high-volume items.

Overall, this dataset offers rich opportunities for building data-driven dashboards, optimizing product and marketing strategies, and enhancing profitability through targeted actions.